Phase 1: Conception Phase

Name: Kandambige Janith Sameera

Matriculation: 10254336

1. Introduction

Online accommodation platforms such as Airbnb allow travelers to search for short term lodging offered by hosts around the world. These platforms connect guests who are looking for temporary accommodation with hosts who want to rent out their properties. The system manages many activities including user registration, listing of properties, searching available rooms, making bookings, processing payments, leaving reviews, and managing communication between users. Because these activities involve multiple users and large volumes of data, a well structured database is required to store and organize the information efficiently.

The purpose of this project is to design a database system for an Airbnb style hotel and room booking platform. The database must support different types of users, manage property and room information, track bookings and payments, and allow guests to provide feedback through reviews and ratings. A properly designed database ensures data consistency, efficient retrieval of information, and smooth functioning of the booking system. In Phase 1 of this project, the focus is on identifying system requirements and designing an Entity Relationship model that represents how different entities in the booking system interact with each other.

2. Requirements Specification

The Airbnb style booking system involves several types of users and functions that interact with the database. The system must support the management of user accounts, property listings, room availability, bookings, payments, and customer feedback. The following sections describe the main user groups and the actions performed within the system.

2.1 Roles and User Groups

The system includes multiple user roles, each responsible for different activities within the platform. The primary role is the guest, who searches for properties, books rooms, makes payments, and provides reviews after the stay. Another important role is the host, who lists properties, manages room availability, sets pricing, and communicates with guests regarding bookings.

The platform also includes an admin role that supervises the entire system. Administrators manage user accounts, verify listings, handle disputes, and ensure that the platform operates according to policies. A payment processor role is responsible for handling secure transactions and managing payment confirmations within the system. In addition, a support agent assists users by resolving complaints, responding to queries, and managing issues related to bookings or payments.

2.2 Actions Performed by Each Role

Each user role performs specific operations within the platform. Guests typically register on the platform by creating an account and providing personal information. After registration, guests can search for available rooms based on location, price, or other preferences. Once a suitable property is found, the guest can book a room for selected dates and complete the payment process. After their stay, guests may submit reviews and ratings to describe their experience. Guests also have the ability to cancel bookings according to the platform's cancellation policies.

Hosts perform actions such as registering their accounts, listing new properties, adding room details, updating availability, and setting prices for bookings. Hosts can also communicate with guests through the messaging system and manage booking requests.

Administrators verify the authenticity of users and property listings and manage system level operations. They may also monitor activities, resolve disputes between guests and hosts, and ensure compliance with platform policies. Payment processors confirm payments and record transaction information, while support agents assist users in solving issues related to bookings, refunds, or account verification.

2.3 Required Data and System Functions

The system must store several types of data in order to support its operations. User profile data includes personal information such as name, contact details, and account credentials. Property data contains information about accommodations listed by hosts, including property type, description, and maximum capacity. Room data provides details about individual rooms, pricing per night, and availability.

The system must also maintain booking records that include check in and check out dates, booking status, and total payment amount. Payment records store transaction details such as payment method, payment date, and payment status. Review and rating data allow guests to provide feedback about their stay and evaluate properties. Additional functions include messaging between users, managing discounts or promotions, tracking cancellations, and storing information about amenities offered by properties. These data elements ensure that the platform can efficiently handle all operations related to accommodation booking and management.

3. ER Model Description

The Entity Relationship model represents the logical structure of the database and illustrates how different entities in the system are connected. In the Airbnb booking system, the central entity is the user, which represents individuals interacting with the platform. A user may take different roles such as guest, host, or administrator.

Hosts are associated with properties that they list on the platform. Each property may contain multiple rooms that guests can book for specific dates. Guests create bookings for rooms, and each booking is associated with payment information that records the transaction details.

Properties are also linked with addresses that specify their location within a particular city and country. Guests may leave reviews for properties after completing their bookings, and each review may include detailed rating scores. Amenities describe the facilities available in properties, while availability calendars record the dates on which rooms are available for booking. Additional relationships exist for cancellations, discounts, wishlists, and messaging between users.

The ER model therefore illustrates how entities such as users, properties, rooms, bookings, payments, and reviews interact with each other. This model provides a clear conceptual representation of the database before implementing the physical tables in the next phase of development.

4. Data Dictionary

The data dictionary provides a structured description of the main entities, their attributes, data types, and meanings within the database system. It helps developers and database designers understand the purpose of each field and ensures consistency in database implementation.

The User entity contains attributes such as user_id which serves as the primary key, first_name and last_name representing the user's name, email and phone for communication, password_hash for account authentication, and created_at which records the account creation time.

The Guest entity includes guest_id as the primary key, user_id as a foreign key linking to the user table, identity_verified which indicates whether the user's identity has been confirmed, and profile_photo which stores the image of the user profile.

The Host entity contains host_id as the primary key, user_id as a foreign key referencing the user account, host_since indicating the date the host joined the platform, response_rate showing how quickly the host responds to messages, and superhost_status which indicates whether the host has achieved a special recognition status.

The Property entity includes property_id as the primary key, host_id as a foreign key linking to the host, title and description describing the property, property_type identifying the accommodation category, and max_guests indicating the maximum number of guests allowed.

The Room entity contains room_id as the primary key, property_id as a foreign key linking to the property, room_type describing the type of room, price_per_night representing the nightly cost, and capacity indicating the number of people the room can accommodate.

The Booking entity includes booking_id as the primary key, guest_id and room_id as foreign keys linking to the guest and room entities, check_in_date and check_out_date indicating the booking period, booking_status showing the current status of the reservation, and total_amount representing the total cost of the booking.

The Payment entity includes payment_id as the primary key, booking_id as a foreign key linking to the booking, payment_method_id indicating the payment method used, payment_date showing when the payment occurred, amount representing the transaction value, and payment_status indicating whether the payment was successful.

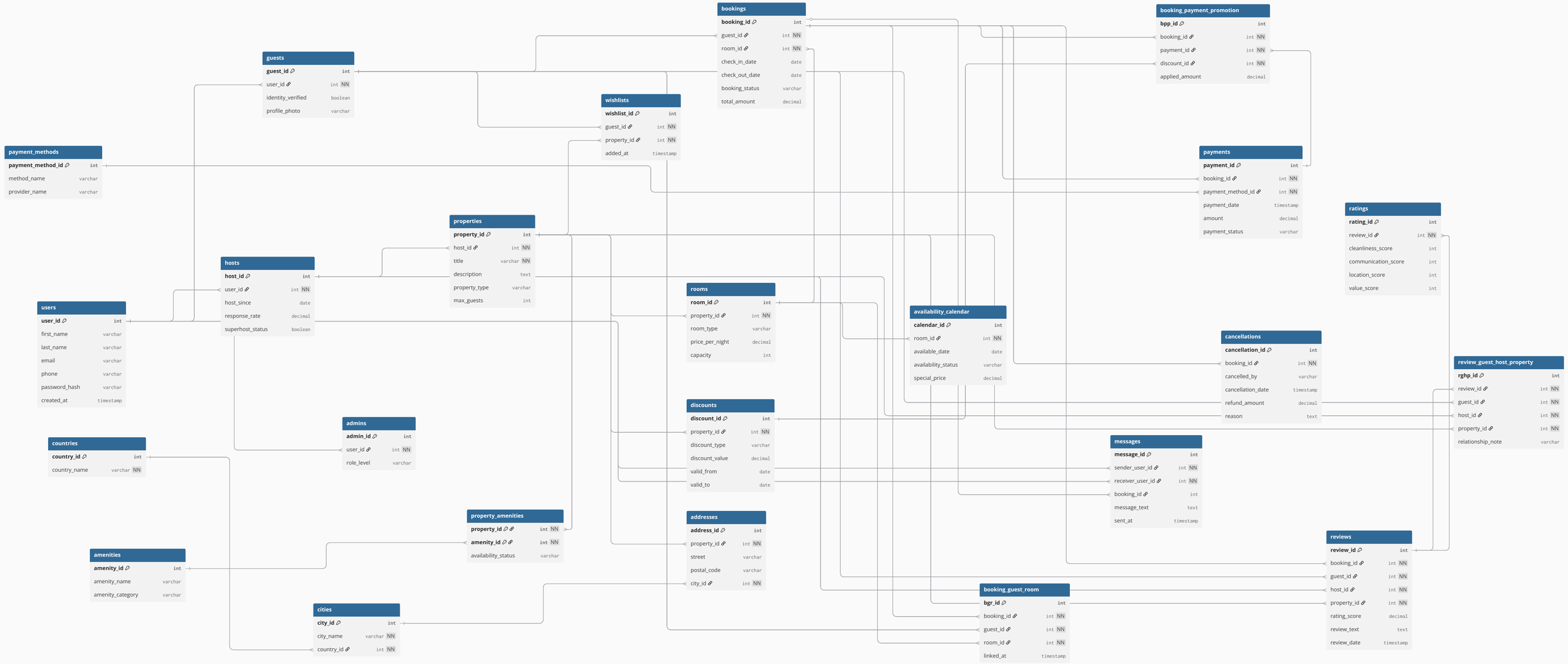
Additional entities such as Review, Rating, Amenity, Availability_Calendar, Cancellation, Message, Discount, and Wishlist store information related to feedback, property features, availability schedules, communication, promotional offers, and saved properties.

5. **Summary**

The rapid growth of online accommodation platforms has increased the need for efficient systems that manage bookings, payments, and user interactions. A platform similar to Airbnb requires a well organized database to ensure that information about users, properties, rooms, and bookings can be stored and accessed reliably. Without a structured database design, it would be difficult to manage large volumes of booking data and maintain consistency across different system operations.

In this project, the problem addressed is the need to design a structured database model that supports the operations of an online accommodation booking system. The solution approach involves identifying the key entities involved in the system and defining their relationships through an Entity Relationship model. The ER model provides a conceptual representation of how data flows within the platform and how different components interact with each other.

During Phase 1, the primary tasks included analyzing the requirements of the booking system, identifying the main user roles and system functionalities, and designing an ER model representing the database structure. A data dictionary was also prepared to describe the attributes and meanings of each entity. This phase establishes a strong conceptual foundation that will guide the implementation of database tables, queries, and testing activities in the next phase of the project.



dbdiagram.io